Pheneon NPU Core v3

PRODUCT DATASHEET

HIGH-PERFORMANCE EDGE AI INFERENCE ENGINE IP BLOCK

Product Overview: The Pheneon NPU Core v3 is a fully verified, synthesizable SystemVerilog IP block engineered for high-throughput edge AI operations. Designed with a 16×16 weight-stationary processing array, integrated vector activation unit, and direct ONNX compiler lowering, it provides a turnkey matrix compute engine for FPGA and ASIC integrations.

1. TECHNICAL SPECIFICATIONS

HARDWARE ARCHITECTURE

- **Systolic Grid:** 16×16 Weight-Stationary Processing Array
- **Data Precision:** INT8 Activation/Weight inputs, 32-bit signed Accumulation
- **Activation Unit:** Vector Processor with BYPASS & ReLU functions
- **Interface Ports:** AXI-Stream streaming interface wrapper

COMPILER & TOOLCHAIN

- **Input Formats:** Native ONNX Model Parser & Lowering Engine
- **Spatial Transforms:** Built-in 2D `im2col` matrix transformation
- **Memory Stimulus:** Automated `.hex` stimulus generator
- **Simulation:** Verilator 5.028 C++ test harness integrated

2. HARDWARE CYCLE BENCHMARKS

Workload / Test Case	Matrix Dimensions	Configuration	Clock Cycles	Numerical Accuracy
GEMM Single-Row Sweep	$M=1, N=32, K=32$	$K_TILES = 1$	206	PASS (0 diffs)
GEMM Single-Row Sweep	$M=1, N=32, K=64$	$K_TILES = 2$	371	PASS (0 diffs)
GEMM Single-Row Sweep	$M=1, N=32, K=128$	$K_TILES = 4$	701	PASS (0 diffs)
Multi-Row Matrix Scaling	$M=2, N=32, K=64$	Weight Reuse Across Rows	715	PASS (0 diffs)
Multi-Row Matrix Scaling	$M=4, N=32, K=128$	Weight Reuse Across Rows	2,723	PASS (0 diffs)
2D Spatial Convolution	$5 \times 5 \times 1$ In, 3×3 Kernel	`im2col` ($M=9, K=32, N=32$)	1,638	PASS (0 diffs)

3. ANALYTICAL PERFORMANCE MODEL

Latency Scaling Equation:

```
Cycles(M, K_TILES) = 206 + (M - 1) × (165 × K_TILES + 12)
```

Weight-stationary execution allows subsequent matrix rows ($M > 1$) to execute with reduced cycle overhead by reusing resident weights.

Commercial Licensing & IP Evaluation

Available for commercial licensing, FPGA customization, and SoC ASIC integration.

Inquire for IP Package